

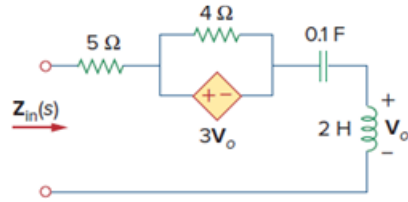
Problem

Refer to the network in Fig. 14.96.

(a) Find $\mathbf{Z}_{in}(s)$.

(b) Scale the elements by $K_m = 10$ and $K_f = 100$. Find $\mathbf{Z}_{in}(s)$ and ω_0 .

Figure 14.96:



Step-by-step solution

Step 1 of 11

Refer to Figure 14.96 in the textbook.

(a)

Insert a 1 V source at the input terminals as shown in Figure 1.

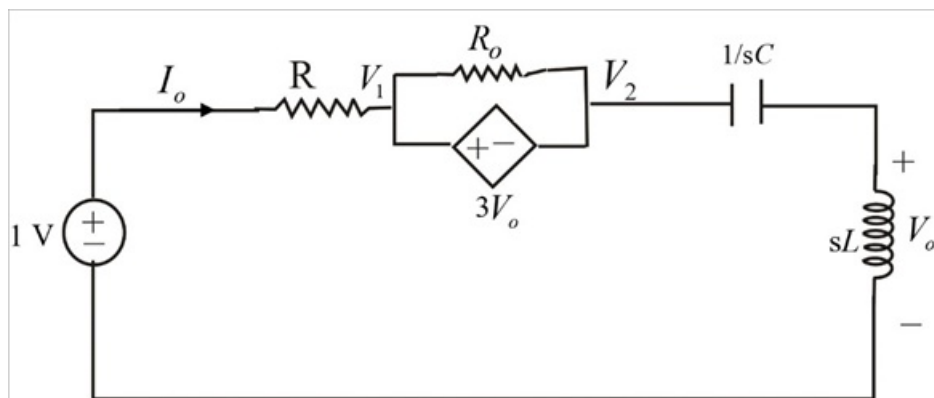


Figure 1

Step 2 of 11

Apply KCL at the super node in Figure 1.

$$\frac{1 - V_1}{R} = \frac{V_2}{sL + \frac{1}{sC}} \quad \dots\dots (1)$$

Also, From Figure 1,

$$V_1 = V_2 + 3V_o$$

$$V_2 = V_1 - 3V_o \quad \dots\dots (2)$$

Step 3 of 11

Now determine the voltage V_o using voltage division rule.

$$V_o = \left(\frac{sL}{sL + \frac{1}{sC}} \right) V_2$$

$$\frac{V_o}{sL} = \left(\frac{V_2}{sL + \frac{1}{sC}} \right) \dots\dots (3)$$

$$V_o = \left(\frac{s^2 LC}{s^2 LC + 1} \right) V_2$$

$$V_2 = \left(\frac{s^2 LC + 1}{s^2 LC} \right) V_o \dots\dots (4)$$

Step 4 of 11

Equate equation (2), and (4).

$$V_1 - 3V_o = \left(\frac{s^2 LC + 1}{s^2 LC} \right) V_o$$

$$V_1 = \left(\frac{s^2 LC + 1}{s^2 LC} + 3 \right) V_o$$

$$V_1 = \left(\frac{4s^2 LC + 1}{s^2 LC} \right) V_o$$

$$V_o = \left(\frac{s^2 LC}{4s^2 LC + 1} \right) V_1 \dots\dots (5)$$

Step 5 of 11

Now substitute $\left(\frac{V_2}{sL + \frac{1}{sC}} \right)$ for $\frac{V_o}{sL}$ from equation (3) in equation (1).

$$\frac{1 - V_1}{R} = \frac{V_2}{sL + \frac{1}{sC}}$$

$$\frac{1 - V_1}{R} = \frac{V_o}{sL}$$

Substitute $V_o = \left(\frac{s^2 LC}{4s^2 LC + 1} \right) V_1$ from equation (5).

$$\frac{1 - V_1}{R} = \frac{\left(\frac{s^2 LC}{4s^2 LC + 1} \right) V_1}{sL}$$

$$\frac{1 - V_1}{R} = \frac{sC V_1}{(4s^2 LC + 1)}$$

$$(4s^2 LC + 1) - (4s^2 LC + 1)V_1 = sRC V_1$$

$$(4s^2 LC + 1) = (sRC + 4s^2 LC + 1)V_1$$

$$V_1 = \left(\frac{4s^2 LC + 1}{sRC + 4s^2 LC + 1} \right) \dots\dots (6)$$

Step 6 of 11

Determine the current I_o .

$$I_o = \frac{1 - V_1}{R}$$

Substitute $\left(\frac{4s^2LC + 1}{sRC + 4s^2LC + 1} \right)$ for V_1 from equation (6).

$$I_o = \frac{1 - \left(\frac{4s^2LC + 1}{sRC + 4s^2LC + 1} \right)}{R}$$

$$I_o = \frac{\left(\frac{sRC + 4s^2LC + 1 - 4s^2LC - 1}{sRC + 4s^2LC + 1} \right)}{R}$$

$$I_o = \frac{\left(\frac{sRC}{sRC + 4s^2LC + 1} \right)}{R}$$

$$I_o = \left(\frac{sC}{sRC + 4s^2LC + 1} \right)$$

Step 7 of 11

Determine the impedance $Z_{in}(s)$.

$$Z_{in} = \frac{1 \text{ V}}{I_o} \quad (\because 1 \text{ V} = \text{Assumed source voltage})$$

Substitute $\left(\frac{sC}{sRC + 4s^2LC + 1} \right)$ for I_o .

$$Z_{in}(s) = \frac{1}{\left(\frac{sC}{sRC + 4s^2LC + 1} \right)}$$

$$Z_{in}(s) = \frac{1 + sRC + 4s^2LC}{sC} \dots\dots (7)$$

Step 8 of 11

Substitute 5Ω for R , 2 H for L , and 0.1 F for C .

$$\begin{aligned} Z_{in}(s) &= \frac{1 + s(5)(0.1) + 4s^2(2)(0.1)}{s(0.1)} \\ &= \frac{1 + 0.5s + 0.8s^2}{0.1s} \\ &= \frac{10 + 5s + 8s^2}{s} \\ &= 8s + 5 + \frac{10}{s} \end{aligned}$$

Therefore, the required impedance is $Z_{in}(s) = 8s + 5 + \frac{10}{s}$.

Step 9 of 11

Determine the resonance frequency ω_o .

$$\omega_o = \frac{1}{2\sqrt{LC}}$$

Substitute 2 H for L , and 0.1 F for C .

$$\begin{aligned}\omega_o &= \frac{1}{2\sqrt{(2)(0.1)}} \\ &= \frac{1}{2\sqrt{(0.2)}} \\ &= 1.118 \text{ rad/s}\end{aligned}$$

Thus, the resonance frequency is $\omega_o = 1.118 \text{ rad/s}$.

Step 10 of 11

(b)

On scaling resistance,

$$\begin{aligned}R' &\rightarrow K_m R \\ 4 \Omega &\rightarrow 10(4) = 40 \Omega \\ 5 \Omega &\rightarrow 10(5) = 50 \Omega\end{aligned}$$

On scaling inductance,

$$\begin{aligned}L' &\rightarrow \left(\frac{K_m}{K_f}\right)L \\ L' &\rightarrow \left(\frac{10}{100}\right)(2) = 0.2 \text{ H}\end{aligned}$$

On scaling capacitance,

$$\begin{aligned}C' &\rightarrow \left(\frac{1}{K_m K_f}\right)C \\ C' &\rightarrow \left(\frac{1}{10(100)}\right)(0.1) = 1 \times 10^{-4} \text{ H}\end{aligned}$$

Determine the scaled value of impedance $Z'_{in}(s)$ using equation (7).

$$Z'_{in}(s) = \frac{1 + sR'C' + 4s^2L'C'}{sC'}$$

Substitute 50Ω for R' , 0.2 H for L' , and $1 \times 10^{-4} \text{ H}$ for C' .

$$\begin{aligned}Z'_{in}(s) &= \frac{1 + s(50)(10^{-4}) + 4s^2(0.2)(10^{-4})}{s(10^{-4})} \\ &= \frac{10^4 + s(50) + 4s^2(0.2)}{s} \\ &= \frac{10^4 + 50s + 0.8s^2}{s} \\ &= 0.8s + 50 + \frac{10^4}{s} \Omega\end{aligned}$$

Therefore, the required scaled value of impedance is $Z'_{in}(s) = 0.8s + 50 + \frac{10^4}{s} \Omega$.

Step 11 of 11

Determine the resonance frequency at scaled values ω'_o .

$$\omega'_o = \frac{1}{2\sqrt{L'C'}}$$

Substitute 0.2 H for L' , and 1×10^{-4} H for C' .

$$\begin{aligned}\omega'_o &= \frac{1}{2\sqrt{(0.2)(10^{-4})}} \\ &= \frac{1}{2 \times 10^{-2} (0.447)} \\ &= \frac{100}{0.8944} \\ &= 111.8 \text{ rad/s}\end{aligned}$$

Therefore, the resonance frequency at scaled value is $\omega'_o = 111.8 \text{ rad/s}$.